NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

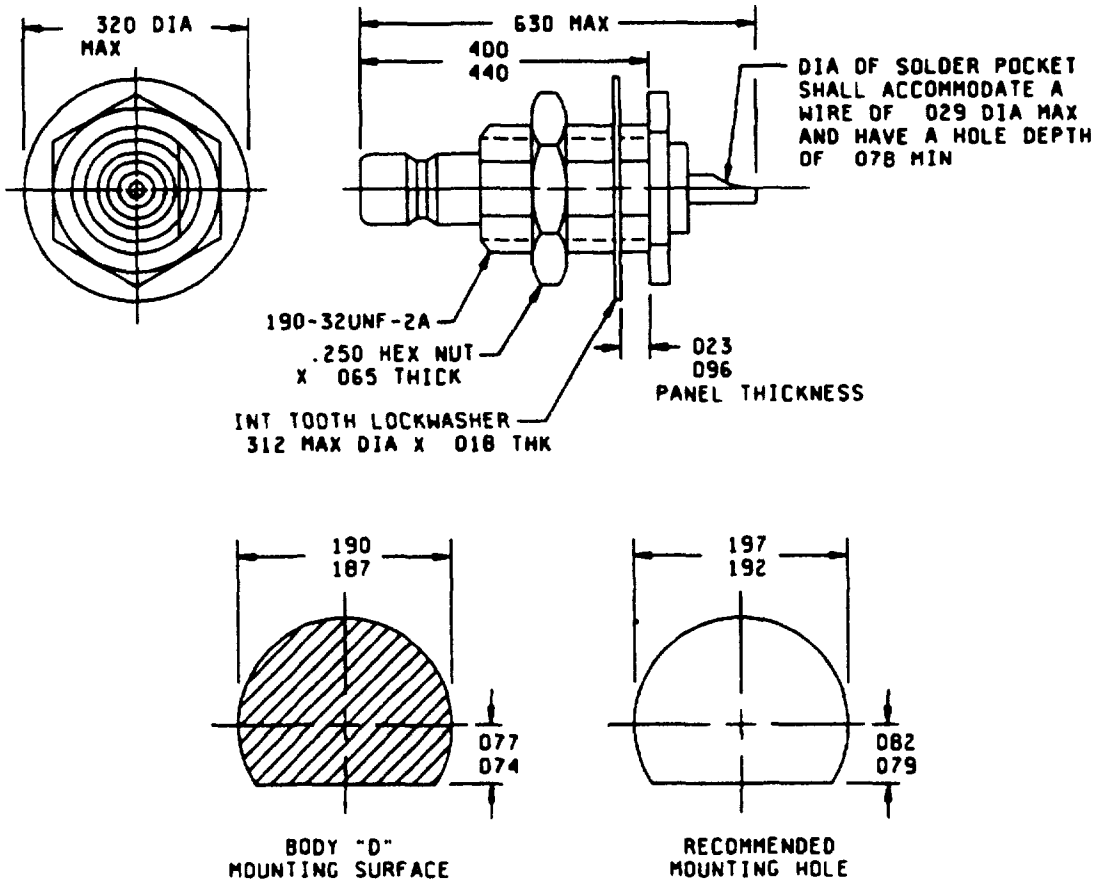
MIL-PRF-39012/71A
24 September 1986
SUPERSEDING
MIL-C-39012/71
22 June 1971

PERFORMANCE SPECIFICATION

CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMB (UNCABLED MALE, JAMNUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the connectors described herein shall consist of this specification sheet and the latest issue of MIL-PRF-39012

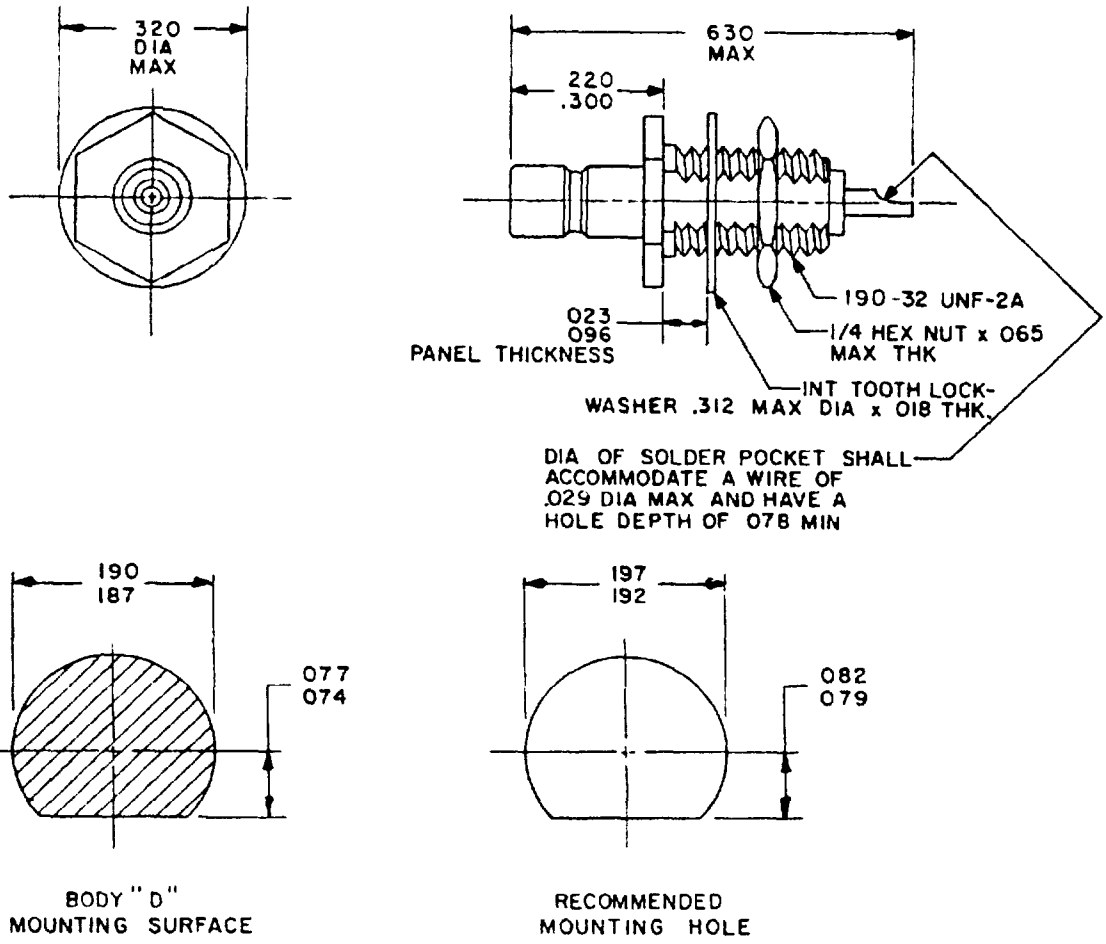


M39012/71-0001 (SEE NOTE 5)
M39012/71-0003

FIGURE 1 General configuration

AMSC N/A
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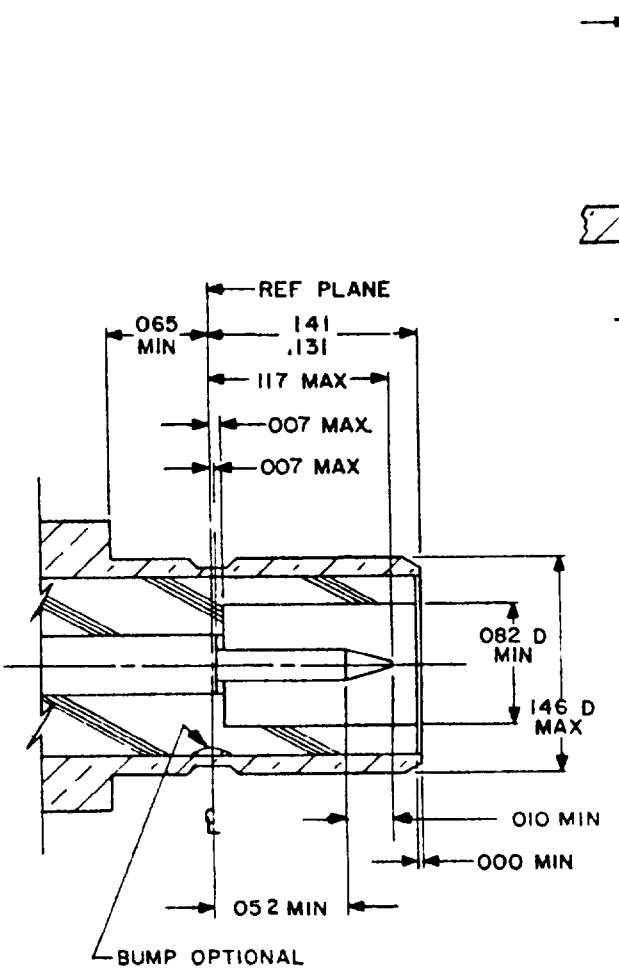
M39012/71-0002 (SEE NOTE 5)
M39012/71-0004

INCHES	MM	INCHES	MM	INCHES	MM
.018	46	.079	2.01	.220	5.59
.023	58	082	2.08	.300	7.62
.029	.74	.096	2.44	.312	7.92
065	1.65	.187	4.75	320	8.13
.074	1.88	.190	4.83	400	10 16
.077	1.96	.192	4.88	.440	11.18
.078	1.98	.197	5 00	.630	16 00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Wrench flats to accommodate standard wrench per FED-STD-H28, appendix 10.
4. All undimensioned pictorial representations are for reference purposes only
5. Connector bodies shall be gold-plated per MIL-G-45204, type II, class 1.

FIGURE 1. General configuration - Continued.

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INCHES	MM	INCHES	MM
.002	.05	.027	.69
.005	.13	.037	.94
.006	.15	.052	1.32
.007	.18	.065	1.65
.010	.25	.082	2.08
.011	.28	.117	2.97
.015	.38	.131	3.33
.019	.48	.141	3.58
.021	.53	.146	3.71

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. All undimensioned pictorial representations are for reference purposes only.

FIGURE 2. Mating dimensions for male terminations.

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ENGINEERING DATA·

Nominal impedance 50 ohms.

Frequency range 0 to 4 GHz.

Voltage rating: 335 V rms maximum at sea level, 85 V rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS·

Dimensions and configuration. See figure 1.

Force to engage and disengage:

Longitudinal force Not applicable.

Torque Not applicable.

Coupling proof torque. Not applicable.

Inspection conditions: Not applicable.

Mating characteristics. See figure 2 for dimensions.

Hermetic seal: Not applicable.

Leakage (pressurized connectors) Not applicable.

Insulation resistance Method 302, test condition B, MIL-STD-202. 1,000 megohms minimum.

Center contact retention:

Minimum axial force. 6.0 pounds from mating end; 4.0 pounds from opposite end.

Torque: 3 inch-ounces.

Resistance to test prod damage· Not applicable.

Corrosion (salt spray). Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR)· Not applicable.

Swept frequency VSWR test setup· Not applicable.

Connector durability

Insertion and withdrawal force: 500 cycles minimum at 12 cycles per minute maximum. The mating force shall meet the mating characteristics requirements.

Initial: 14 pounds maximum.

Final· 14 pounds maximum engage and disengage, 2 pounds minimum disengagement.

Contact resistance In milliohms maximum

	Initial	After environment
Center contact	6.0	8.0
Outer contact·	1.0	1.5
Braid to body	Not applicable	Not applicable

Dielectric withstanding voltage Method 301 of MIL-STD-202. 1,000 V rms at sea level.

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Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Shock (specified pulse): Method 213, test condition B, MIL-STD-202.

Thermal shock Method 107, test condition B, MIL-STD-202

Moisture resistance: Not applicable.

Corona level: Not applicable.

RF high potential withstanding voltage·

Voltage and frequency· 600 V rms at 5 MHz.

Leakage current: Not applicable.

Cable retention force: Not applicable.

Coupling mechanism retention force: Not applicable.

RF leakage· Not applicable.

Insertion loss: Not applicable.

Part number: M39012/71- (dash number from table I).

Group qualification· See table II.

TABLE I. Part numbers.

Part number 39012/71-	Type of mount
0001, 0003	Rear
0002, 0004	Front

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
I	M39012/71-0001 M39012/71-0002	M39012/71-0001, M39012/71-0003 1/ M39012/71-0002, M39012/71-0004 1/

1/ Corrosion (salt spray) and contact resistance data must be submitted to DESC-EQP before qualification approval may be granted.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - EA, MI
Navy - OS
Air Force - 11, 17
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3553-5)

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